

IoT Based Vehicle Age Monitoring System

Project Description

Developed an IoT-based Vehicle Age Monitoring System using NodeMCU, DHT11, BMP180, and ultrasonic sensors to track vehicle condition in real time. The system monitors temperature, pressure, and distance, sends alerts, and improves vehicle safety, maintenance, and performance.

Components Used

Component	Purpose
NodeMCU ESP8266	Microcontroller with WiFi support
DHT11 Sensor	Temperature and Humidity Monitoring
BMP180 Sensor	Pressure Monitoring
Ultrasonic Sensor	Obstacle Distance Detection

Arduino Source Code

```
// IoT Based Vehicle Age Monitoring System
// Components:
// NodeMCU ESP8266
// DHT11 Temperature Sensor
// BMP180 Pressure Sensor
// Ultrasonic Sensor

#include <ESP8266WiFi.h>
#include <DHT.h>
#include <Wire.h>
#include <Adafruit_BMP085.h>

// WiFi Credentials
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// DHT11 Setup
#define DHTPIN D4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// Ultrasonic Sensor Pins
#define TRIG D5
#define ECHO D6

// BMP180 Setup
Adafruit_BMP085 bmp;

void setup() {

  Serial.begin(9600);

  // DHT11 Start
  dht.begin();

  // Ultrasonic Sensor
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  // BMP180 Start
```

```

    if (!bmp.begin()) {
        Serial.println("BMP180 Sensor Not Found!");
        while (1);
    }

    // WiFi Connection
    WiFi.begin(ssid, password);

    Serial.println("Connecting to WiFi...");

    while (WiFi.status() != WL_CONNECTED) {
        delay(1000);
        Serial.print(".");
    }

    Serial.println("");
    Serial.println("WiFi Connected");
}

void loop() {

    // ----- DHT11 -----
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    // ----- BMP180 -----
    float pressure = bmp.readPressure() / 100.0; // hPa

    // ----- Ultrasonic Sensor -----
    long duration;
    float distance;

    digitalWrite(TRIG, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG, LOW);

    duration = pulseIn(ECHO, HIGH);

    distance = duration * 0.034 / 2;

    // ----- Serial Monitor Output -----
    Serial.println("==== Vehicle Monitoring Data =====");

    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" °C");

    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");

    Serial.print("Pressure: ");
    Serial.print(pressure);
    Serial.println(" hPa");

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    // ----- Alert Conditions -----

    if (temperature > 40) {
        Serial.println("ALERT: High Temperature!");
    }

    if (pressure < 900) {
        Serial.println("ALERT: Low Pressure!");
    }

    if (distance < 20) {
        Serial.println("ALERT: Obstacle Detected!");
    }

    Serial.println("-----");

    delay(3000);
}

```

